Where to start?

Version 1.2 by Tom N4FWD May2026

One of the major Pluses for MeshCore is the ability to operate completely off-grid. No fees to a carrier (like AT&T as an example), text messages can be sent through encrypted channels legally (think emergency or private communications and such).

MeshCore vs Meshtastic

Both are built around LoRa (Long Range) radio technology. Link >> LoRa

Comparisons: Link >> Compare

MeshCore basics

Two different camps of MeshCore: Link >> The Split

The Andy camp MeshCore: Link >> https://meshcore.co.uk/

The developers camp MeshCore: Link >> https://meshcore.io/

Another MeshCore site: Link >> https://lowmesh.com/

I chose https://meshcore.io/ as their firmware is more refined. The main developer has this site for obtaining firmware extras: https://buymeacoffee.com/ripplebiz/extras

Apologies, I'm getting ahead of myself. Where can one get equipment to join a MeshCore network?

Note: The Etsy links are for manufactured products. The other links includes Maker style development kits as well as manufactured products.

Places to purchase MeshCore equipment

Amazon: Link >> Amazon - MeshCore query

Etsy – PeakMesh vendor: Link >> PeakMesh

Etsy – Zacorp vendor: Link >> Zacorp

Etsy – General search: Link >> General search for MeshCore

NoDakMesh: Link >> NoDakMesh

Seeed Studio: Link >> Seeed Studio

Any many more which can be discovered via a Internet search.

MeshCore app for Android Phone

Either Google Play Store ***or*** **meshcore/com.liamcottle.meshcore.android**

The app is also available on the Apple Store!

WARNING: The Phone app will require a Companion Device. Also, leaving your phone app connected to your companion device for long periods (many hours) can cause a significant drain on the phone's battery!

You've purchased a device, Next Step – firmware installation or update

Go to the main website https://meshcore.io/ and click on the **FLASHER** link at the top of the webpage. Find your device in the list. When you hover over the title, an image of the device will pop up. Click on the title and you will see the various roles your device can do. *For a T-Deck, you can install the **Ripple Firmware**. With a SD card installed, you can use the **Ripple GUI: data on SD card** version.*

Community Firmware Roles (can vary depending on the device):

Companion Bluetooth:

This role is one where you have the MeshCore app on your phone or computer and wish to connect to you device using Bluetooth Low Energy (BLE). In this role, your device will act similar to a translating modem, basically enabling Smart Phone to LoRa connectivity.

Companion USB:

This role is similar to the previous role except you are using a USB cable between the software app and the device. Example: *Laptop to LoRa connectivity.*

Repeater:

In this role, your device will function as a repeater, similar to the Amateur Radio repeaters.

Room Server:

As a room server, your device will act similar to an old style BBS where incoming messages are stored until retrieved by the recipient of the message. Why use something like this? MeshCore (and Meshtastic) are live messaging modes of operation. If a recipient turns off their device, or goes out of range of the MeshCore network, or the battery dies, the room server will automatically capture and hold the message(s) until the recipient can retrieve the messages.

KISS Radio Modem:

Using USB cabling, the device can link up the LoRa network to Amateur Radio *packet radio* software. As far as I know, this role is experimental.

Of all of the roles available, you will most likely be using the first & third roles.

Flashing

Note: All flashing of firmware uses a USB cable between your computer and the device.

When you have connected your device via USB and click on the ***Flash!*** Button, you will need to select the serial port which the flashing software will use to communicate with the device. Just select as prompted by the flash software. When the flashing is finished, you can power-cycle your device to reboot it with the new firmware.

Lilygo T-Deck (a self contained device)

For my texting devices, I chose the Lilygo T-Deck Plus (available from Amazon). It has a touch screen, trackball as well as a small keyboard (which takes a bit of getting used to). *Use the trackball to scroll through menus / listings. The trackball looks like a button in the center, right above the keyboard.*

The T-Deck has a slot for a micro-SD card. If you decide to add a micro-SD card, use either a 16Gb or 32Gb formatted to FAT32 (or VFAT in Linux - different name, same format).

In the Ripple GUI on a T-Deck, tap in the lower left corner (the hamburger icon) and tap on the **Networks** icon. Then tap the upper right and select **New Network**. Select **MeshCore** and press **Enter**. Scroll down to the **US/Canada (Recommended)** and press **Enter**. Use the **Backspace** key and rename the network. Remember to press **Enter** when finished. The network name should reflect either an existing region which you can join or the actual area where you live. Keep in mind the network can grow.

Once you are done, the network should show in the contact list. Select the network and tap on the **Identity** icon. Use the **Backspace** key and rename your device. Then press **Enter**.

More Info, but **older**: Link >> https://buymeacoffee.com/ripplebiz/ultra-v7-7-guide-meshcore-users

Map Function

The micro-SD card can store the maps used in the T-Deck map function. The maps file can be purchased here: https://buymeacoffee.com/ripplebiz/extras When the maps file is unzipped, the **tiles** subdirectory and all of its contents will need to be copied to the root directory of the micro-SD card. So unzip the file on your computer and then copy over only the **tiles** sub-directory to the micro-SD card.

Launcher (for T-Deck)

With a micro-SD card installed, you can take advantage of a launcher program: https://github.com/bmorcelli/Launcher

To take advantage of the launcher, you will need to store the firmware binaries in the root directory of the micro-SD card. When you are getting ready to flash your firmware, in the lower right corner of the flasher page, you will see a **Download** button. Select that and then the simple binary file, **not the *merged* binary.** Store the downloaded binary where you can find it.

In the case of a T-Deck, download the SD card version of the firmware.

Once you have the binary file downloaded, copy it to the root directory of the micro-SD card. The launcher software will allow you to select it for installation. Due to the limited storage of the micro-controller of the LoRa board, the launcher will have to reload the selected firmware.

Not to worry, the launcher program does not interfere with the configuration(s) of firmware. That is to say, for example, if you unlock the Ripple GUI, you will not need to do that again.

More gotchas, though minor

Unlock Key

Each firmware, which you can load onto a T-Deck, is free for the basic functionality. However, to get the extended functionality, you will need to pay a one-time low cost fee for the unlock key. A purchased unlock key is good forever, but is assigned to the serial number of the device. Meaning, you will need another unlock key if you purchase another T-Deck.

Maps

The maps mentioned earlier in this document are a different extended function. Fortunately, once you purchase the maps file, you can install it on any additional T-Decks without having to purchase maps file again. You will need to unlock the GUI to access the maps in the GUI.

GPS

In order to keep the messaging in proper time order, for devices with GPS capability, a GPS satellite lock is necessary. So keep that in mind if you happen to be in a building with metal construction.

Downsides

One downside is that the devices are battery operated. That situation can be mitigated by using solar cells power like found in a dedicated repeater.

A second downside is the necessity of having multiple LoRa repeaters who can see other LoRa repeaters. A robust MeshCore mesh will require many LoRa repeaters because of the 33cm band (915 MHz) frequency. Read the example on this page (scroll down to the Feature area): https://www.seeedstudio.com/MeshCore-Starter-Kit-Ready-to-Use-Off-Grid-Instant-Reliable-Communication.html

Private Channels

To distribute a private channel on *MeshCore*, you must manually share the unique secret key with trusted users, as private channels do not derive keys from channel names. This is typically done via QR codes (*printed on a card for distribution*).

Or direct copy-paste, ensuring the key is exchanged through a *secure, out-of-band method.* (**Do not trust a US based email services for sending the information as they spy on your emails. Use a privacy email service like these:** Privacy oriented email services *I use protonmail.com*)

Lilygo T-Deck - Running Ripple GUI Firmware 10.0 (or higher)

Creating a Private Channel

Step 1) Tap the L in the left side column and tap the network (at the top of the listing)

Step 2) Tap the icon for the first unused channel, typically Channel 2. Channel 1 is the public channel.

Step 3) Type in the name of the new private channel. Using the scroll button of the T-Deck, scroll down to the **Hex** option. Tap the hamburger icon in the upper right corner and tap on **New 128 Bit Key**. Then press the **Enter** key to save the settings.

Sharing a newly created private channel to a T-Deck

Have a print out of the key to be used for the new private channel handy. Follow the above instructions and give the new private channel the same name as on the originating device, **however**, instead of generating a new key, type in the key from the print out. Then press **Enter** to save the channel.

Once the channel has been saved, select it in your contact list and send a message. All other connected devices on the MeshCore network with the same private channel should receive the message on that private channel.

Sensors

The MeshCore supports sensors like a weather station as an example.

From the MeshCore Discord contributor **Mike's Allotment**:

The reason I don't want to use a 'pull' model is because the Sensor nodes have to be active/alive all the time listening for pull requests. This eats into battery life.

The push model means that the sensors can be in deep sleep for 99% of the time and only active for a short time when they wake up, take a temp/humidity reading and push a packet out. This hugely extends battery life.

And

A better information page: Link >> https://meshcore-dev.github.io/meshcore-ha/docs/ha/sensors/

Or this site: Link >> https://github.com/netmilk/esphome-meshcore

General Information

Link >> https://www.regionmesh.com/

Adding your device to the network

After flashing your device and configuring the name, set the unit to the **Discover Channel** and send a "**advert**". Adverts allow you advertise your device to the network. Any listening nodes within range of the MeshCore network will get your advent and then can add your device to their contact listing.

To send an **advert** from your device, scroll to the **Discover Channel** and select it, tap the upper right icon and select **Send ID Broadcast**.

MeshCore Network detection

Repeaters are usually set up to automatically send an advert at discrete intervals. Go to the map at https://map.meshcore.io/ and navigate to your area. If you see any of the shown repeaters sending **adverts** in your **Discover Channel,** then you are potentially within the coverage area. Switch to the **Public Channel** and send a text announcing yourself. If your text was received by a repeater, you will see a number next to your message. Otherwise, you will see a timeout error appear. Devices receiving the message can then add your node to their contact list.